

School of Computer Science Engineering and Information Systems
Fall Semester 2026-2027

BAITE203- Computer Networks and Data Communications

Faculty: Dr. G. Usha Devi

Case Study: University Campus Network Design

Scenario

A university has three academic blocks, one library, one administration building, and two hostels. More than **2,500 users** access the network simultaneously. Students frequently complain about slow Internet access, while faculty require uninterrupted access to online learning platforms and research databases.

Problem

Design a scalable, secure, and high-speed campus network.

Tasks

- Design the network topology.
- Select suitable switches and routers.
- Divide the network using VLANs.
- Recommend IP addressing scheme.
- Suggest routing protocol.
- Propose security measures.

=====